Check-In 08/24. *(True/False)* If $f(x)$ is a relation with $f(1) = 8$, $f(2) = 8$, and $f(3) = 3$, then $f(x)$ cannot be a function.

Solution. The statement is *false*. Recall a function is a relation such that for each input, there is only one possible output (though that output can be repeated by other inputs). That is, we can think of a function as a 'coherent set of instructions.' Given an input, is there a single, unambiguous output? We know $f(1) = 8$, $f(2) = 8$, and $f(3) = 3$. There is no confusion as to what the output for $f(x)$ is for $x = 1, 2, 3$. In fact, $f(x) = -\frac{5}{2}x^2 + \frac{15}{2}x + 3$ is a function with $f(1) = 8$, $f(2) = 8$, and $f(3) = 3$.

Check-In 08/26. *(True/False)* The function $\ell(x) = 9 \, - \, 4x$ is linear, has y-intercept 9, has slope 4, and is an increasing function.

Solution. The statement is *false*. We know a linear function has the form $y = mx + b$. Observe that $\ell(x)$ has the form $mx + b$ with $m = -4$ and $b = 9$. Therefore, $\ell(x)$ is linear. We know the slope is m, and $m = -4$ (not 4), and the y-intercept is b, and $b = 9$. We also know the y-intercept occurs when the input is 0, i.e. $x = 0$. But we know $\ell(0) = 9 - 4(0) = 9 - 0 = 9$. Finally, we know a linear function is increasing if $m > 0$, constant if $m = 0$, and decreasing if $m < 0$. Because $\ell(x)$ has $m = -4 < 0$, we know that $\ell(x)$ is decreasing.